

Temporal scale mediates the roles of abundance and traits in plant-pollinator interactions

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Abstract

Flowering plants exhibit an astonishing diversity of reproductive structures, which shapes how they interact with their pollinators. However, to what extent reproductive traits or other drivers like phenology and species abundances shape plant-pollinator interactions is still an open question. To disentangle the relative effects of species phenology, abundances and traits in plant-pollinator interactions, we compared pollinator visitation rates, diet breadth, specialization from plant species that vary in reproductive traits, phenology and abundances. To do this, we observed plant-pollinator interactions for a whole flowering season in three different botanical gardens located in an urban setting. We anticipate that the ecological role of the different plant and pollinator species are largely driven by their phenological overlap and abundance, and only certain taxonomic groups are strongly influenced by trait matching processes. Our results highlight the relevance of considering multiple mechanisms to understand plant-pollinator interactions.

Introduction

Flowering plants, pollinators, and their interactions form complex networks that are fundamental to the architecture of biodiversity. The structure and composition of these networks exhibit strong spatio-temporal variation, shaped by species attributes and environmental conditions (Burkle & Alarcón, 2011; Tylianakis & Morris, 2017). While ecological networks provide insights into the processes that govern community assembly (Bascompte & Jordano, 2007; Guimaraes Jr, 2020), documenting interactions remains extremely labour-intensive (Chacoff et al., 2012). Moreover, the resulting network structures are highly dependent on sampling effort and methodology (Jordano, 2016). At the same time, interactions are often summarised into single metrics and analysed with a diverse array of statistical methods, each offering potentially distinct perspectives on ecological systems (Delmas et al., 2019; Peralta et al., 2024). Despite pronounced temporal dynamics, networks are often treated as static entities (CaraDonna et al., 2017; Poisot et al., 2015), requiring the aggregation or simplification of relevant information when evaluating ecological patterns. The rapid turnover of species and interactions across the season, together with the aggregation of interactions into static network representations, can obscure rather than clarify the processes structuring ecological systems (Brimacombe et al., 2022; Payrató-Borràs et al., 2024; Trøjelsgaard & Olesen, 2016). Addressing these challenges requires systematic approaches that integrate multiple ecological drivers and layers of spatiotemporal complexity to improve both understanding and predictive power in plant-pollinator interactions (Blüthgen & Staab, 2024; Dormann et al., 2017).

There is strong evidence that species phenology, abundances, and traits are key ecological drivers shaping species interactions and network structure (Kaiser-Bunbury et al., 2014; Vázquez et al., 2009), yet assessing their combined influence across spatiotemporal scales remains a major challenge (Olito & Fox, 2015; ?; ?). Phenology is considered one of the main determinants of network structure, as temporal overlap between species constrains the pool of potential interactions (Benadi et al., 2014; Encinas-Viso et al., 2012), with approximately one-third of all potential interactions estimated to be impossible due to phenological mismatches (Olesen et al., 2011). When two species overlap in time, the probability that they interact is generally considered a function of their relative abundances and species traits (Olito & Fox, 2015; Stang et al., 2007), along with other abiotic and biotic factors (e.g. temperature and species competition). While abundance influences both species' roles (Fort et al., 2016; Simmons et al., 2019) and overall network structure (Canard et al., 2014; Vázquez et al., 2009), independent estimates are rarely available and are often inferred from interaction data (Blüthgen & Staab, 2024), potentially obscuring its true contribution to interaction patterns.

Traits, in turn, provide mechanistic insights, as they constrain the range of possible partners and help define species' ecological roles within networks (Bartomeus et al., 2016; Coux et al., 2016; Dehling et al., 2016). For instance, interaction probability often peaks when pollinator proboscis length matches floral tube depth (Klumpers et al., 2019), yet imperfect matches can still allow interactions. Although multiple traits often act simultaneously, capturing their multidimensional nature remains challenging (Eklöf et al., 2013; Lanuza et al., 2023), though

phylogenetic approaches may help integrate such complexity. Further, variation in pollinator generalism may modulate the importance of traits, which is expected to decline with increasing generalism (Peralta et al., 2020). The strong context-dependency of these different ecological drivers constrains predictive power and highlights the need for high-quality data to evaluate multiple scenarios across taxonomic groups and spatiotemporal scales.

Recent evidence suggests that distinct temporal scales can provide different views of the structure and processes that shape plant-pollinator interactions (Miele et al., 2020; Schwarz et al., 2020). Interactions are highly dynamic, shifting markedly between years (Chacoff et al., 2018; Petanidou et al., 2008), across seasons (Souza et al., 2018; Vizentin-Bugoni et al., 2016), and even within days (Schwarz et al., 2021). At fine scales, interactions are shaped by physiological responses to abiotic factors (e.g., temperature, light, precipitation) that determine activity periods, whereas variation across days or weeks also reflects phenological succession and changes in the species pool (Bramon Mora et al., 2020; CaraDonna et al., 2017). Although some of this variation arises from natural dynamics, global change is expected to disrupt and intensify temporal fluctuations (Duchenne et al., 2020; Kharouba et al., 2018). While aggregated snapshots of communities are valuable for describing structural and ecological properties at broad scales (Bascompte & Jordano, 2007), the relative impact of ecological drivers may vary with temporal scale, underscoring the need to account for temporal dynamics to fully understand plant-pollinator interactions. Together, these approaches provide complementary perspectives on the mechanisms structuring ecological interactions and their predictive power.

Predicting species interactions is a central goal in ecology (Peralta et al., 2024; Valdovinos, 2019), particularly in the context of climate change, ongoing species loss, and the need to sustain ecosystem services. However, although overall network structure can often be partially reproduced (Crea et al., 2016; Eklöf et al., 2013), accurately predicting pairwise interactions remains challenging (Peralta et al., 2024; Valdovinos, 2019). Recent efforts to improve predictive performance have sought to address limitations of conventional analytical methods, such as the non-independence of observations and nonlinear effects of key variables, while incorporating ecological drivers such as abundance, traits and phenology (Bartomeus et al., 2016; Benadi et al., 2022). In parallel, flexible machine learning approaches have emerged as promising tools for improving predictions of ecological interactions (Strydom et al., 2021). For example, Pichler et al. (2020) showed that machine learning models outperformed traditional regression approaches when predicting plant-pollinator interactions. Nevertheless, skepticism remains regarding their general applicability, and comparative approaches evaluating predictive performance across different contexts are still needed.

In this study, we evaluate how abundance, trait matching and phenology shape plant-pollinator interactions through time using complementary approaches that assess network structure, driver effects on visitation and predictive performance, alongside null models testing for pollinator preferences beyond phenology and abundance (Figure 1). To achieve this, we collected data on plant-pollinator interactions, along with associated ecological drivers, in three botanical gardens. This setting allows us to test multiple potential combinations of pairwise interactions in a semi-controlled environment that would otherwise be impossible to observe in

natural communities, while also supporting detailed measurement of the different ecological drivers. By integrating these drivers across temporal scales and analytical frameworks, our study provides a mechanistic assessment of the processes shaping plant-pollinator interactions over time and evaluates their potential for predicting ecological networks.

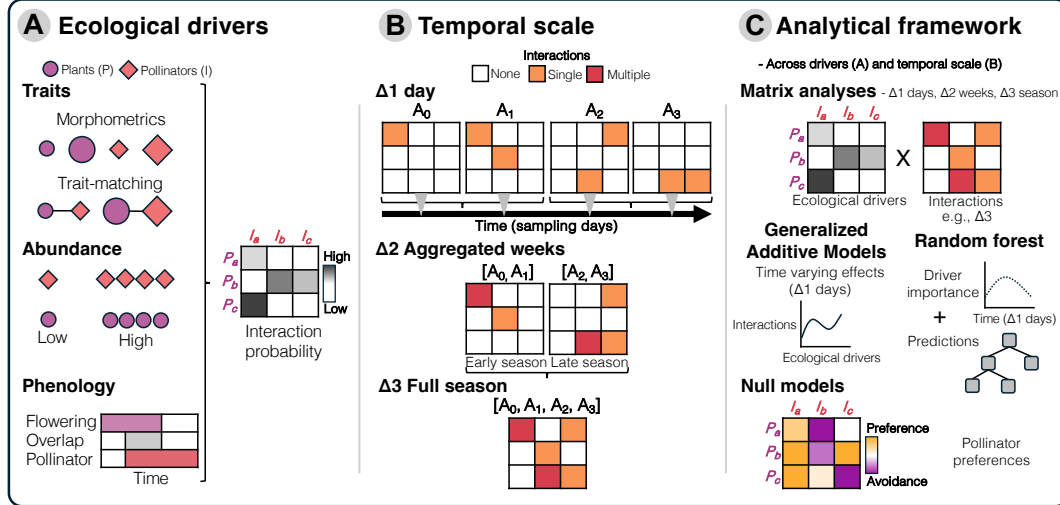


Figure 1. Workflow of the study, showing how different ecological drivers (A) are tested across various temporal scales (B) using different analytical methods (C). Panel A illustrates the main ecological drivers used in this study to evaluate plant-pollinator interaction patterns: (1) pollinator and floral traits, (2) pollinator and floral abundances, and (3) pollinator and floral phenology. Panel B presents the three levels of temporal aggregation considered in this study: (1) individual sampling days, (2) early, mid, and late season, and (3) the full flowering season. Panel C depicts the three main analytical approaches implemented in this study to evaluate the effects of different ecological drivers across distinct levels of temporal aggregation: (1) matrix analyses, (2) generalized linear models (GLMs), and (3) machine learning

Methods

Study site and sampling method

Plant-pollinator interactions were monitored during the entire flowering season of 2023 in the botanical gardens of Halle, Jena, and Leipzig (Germany; Figure S1). These gardens belong to the PhenObs initiative (Nordt et al., 2021), a network of botanical gardens across the Northern Hemisphere that records phenological observations on a wide range of herbaceous species. The objective of PhenObs is to evaluate the impacts of climate change on species performance (e.g., Sporbett et al., 2022). In this study, plant-pollinator interactions were recorded for a subset of species from the PhenObs project within these three gardens.

Each garden was visited weekly, and plant-pollinator interactions were sampled for 15 minutes on each focal plant. On each sampling day, focal plants were surveyed in random order, and only pollinator individuals contacting the reproductive organs of the plant were recorded as interacting species. In addition, to create a comprehensive inventory of pollinators and their interactions within the gardens, ten random sites were selected each sampling day, and plant-pollinator interactions were monitored for three minutes at each site, resulting in a total of 30 minutes of random observations per sampling day and garden.

Pollinator monitoring concluded once PhenObs species had largely ceased flowering in each garden, corresponding to the late phase of the flowering season. Random surveys nevertheless showed continued pollinator activity and flowering of non-PhenObs plants (Figure S2), indicating that sampling was constrained by focal plant phenology rather than pollinator availability. Consequently, two complementary sampling approaches were implemented: targeted observations on PhenObs plants, representing a subset of the available floral resources, and random stops within the gardens designed to characterise the broader pollinator assemblage that may not be captured through this subset.

Sampling effort and taxonomic coverage This study monitored 78 focal plant species from the PhenObs initiative, encompassing 68 genera, 33 families, and 22 orders. Pollinator observations were conducted on 46 plant species in Halle (52.3 h), 43 in Jena (41.3 h), and 47 in Leipzig (49.8 h), with only 12 species shared among all gardens. Complementary random census sampling expanded plant coverage, recording plant-pollinator interactions on 134, 124, and 125 plant species in Halle, Jena, and Leipzig, respectively (10.1, 11.3, and 10.4 h per garden), collectively capturing a broad taxonomic representation of flowering plants. When projected onto the global angiosperm reproductive trait spectrum (?), the PhenObs species included in this study cluster within the region associated with pollinator-attraction traits, indicating that the sampled plant pool is well positioned to investigate the processes mediating plant-pollinator interactions (Figure S3). Focal observations captured a total of 175 pollinator species across the three gardens, with 123 species in Halle, 86 in Jena, and 72 in Leipzig. Random census sampling further enriched pollinator diversity, documenting an additional 68 species not observed on focal plants. Overall, the combined sampling approaches provided a comprehensive overview of the pollinator assemblages within these botanical gardens. Accumulation curves (Figure S4) indicate that sampling effort was sufficient to characterize pollinator assemblages associated with PhenObs plants across gardens. However, random observations proved more efficient at capturing overall pollinator diversity at the garden scale, providing a more comprehensive representation of the pollinator assemblage with lower sampling effort while covering a broader range of plant species. **Data description**

To evaluate the different mechanisms that drive plant-pollinator interactions, detailed information on abundance, phenology and species traits for both plants and pollinators was recorded as follows:

- (i) Floral abundance was annotated for each focal plant species. Additionally, the semi-controlled conditions of botanical gardens facilitated the direct annotation of the number

of pollinator individuals on each sampled plant. Despite its limitations, particularly at higher activity levels when individuals are difficult to track, this approach offers a more realistic estimate of pollinator abundance than the standard method of estimating abundances from interactions, which can bias our understanding of the mechanisms driving network structure (Blüthgen et al., 2006; Vizentin-Bugoni et al., 2014). Note that flowering species with tiny densely packed flowers were considered at inflorescence level (15.2% of the total flowering species considered in this study).

- (ii) Given that plant phenology was already monitored on the PhenObs project, we have weekly phenological records with flowering percent for all focal plants. For 10 species with missing phenology data, we conducted generative additive models (GAM) with our records of flower number to build the phenology on the same sampling dates as the other species by using the mgcv package. Unlike plants, pollinator phenology relies on ‘stochastic’ interaction observations that may not represent their complete flying period. For this reason, we opted to complement our observational data from the Global Biodiversity Information Facility (GBIF). Germany is one of the countries with highest records of occurrence data (Rocha-Ortega et al., 2021), and in particular, their cities (e.g., botanical gardens) contain most of those records (Sweet et al., 2022). Thus, we opted to extract occurrence records around the three cities for the different pollinator species to complement the different pollinator phenologies. We first created a centroid within the 3 cities, and then, created a rectangular shape around it that expanded approximately 1000 kms in longitude and 500 kms in latitude. We used this shape because we expect environmental variables to be more consistent and phenology to be more similar across a longitudinal range than a latitudinal one. Thus, our download resulted in 2,933,772 occurrence records (<https://doi.org/10.15468/dl.k5cnuf>). Pollinator phenology was therefore shared across the three gardens; although this approach has limitations, it provides a more robust estimate than those based solely on interaction networks.
- (iii) Plant traits were selected following the main traits driving the trait spectrum of flowering plants as shown in Lanuza et al. (2023) (i.e., flower number, plant height, style length, flower size, ovule number and selfing). In addition, we included phenological traits (i.e., flower lifespan and flowering duration), floral rewards (i.e., nectar volume and pollen production per flower) and seed weight as a measure of reproductive investment (Table S1a). These traits were used to characterise the position of the studied plant species within the broader reproductive trait spectrum (?), whereas a subset of traits relevant to interaction mechanisms was used in the subsequent analyses. Pollinator traits included intertegular distance, body length and proboscis length (Table S1b), which are key morphological traits mediating plant-pollinator interactions (Földesi et al., 2021; Klumpers et al., 2019). Both intertegular distance and body length were measured from images using ImageJ, and proboscis length was extrapolated for bee species by using the pollimetry package (Kendall et al., 2019). Non-bee species were complemented with proboscis values from the literature and images when possible.

Analysess

To unravel the role of time and analytical methods in our understanding of the ecological drivers of plant-pollinator networks, this study analyses the role of abundance, phenology and traits at three different levels of temporal complexity (i.e., weekly, weekly aggregated and full season) using three different analytical approaches (see **Figure 1**). The three analytical approaches implemented are: (i) matrix analyses using Procrustes tests, (ii) generalized linear models (GLMs), and (iii) machine learning. Each of these methods provides complementary insights into the mechanisms driving plant-pollinator interactions across different temporal scales.

Matrix analyses

Visitation rate networks were built for each botanical garden for pollinators identified to species, and the main pollinator orders (Hymenoptera, Diptera, Coleoptera, Lepidoptera). Plant-pollinator interaction records were aggregated by garden and species pair to generate weighted bipartite interaction matrices. Interaction frequency (visitation rate) networks were calculated by dividing total interactions for each plant-pollinator pair by the total observation time for each plant species within a garden, thereby standardizing visitation rates. These standardized rates were summed across sampling occasions to produce garden-level visitation-rate networks representing cumulative seasonal interaction intensity while accounting for plant-specific sampling effort.

Abundance probability matrices were generated for each botanical garden using standardized relative abundances of pollinator species and floral resources. Pollinator and floral abundances were summed within each garden and converted to relative abundances by dividing species-level abundances by total community abundance. Finally, interaction probabilities were calculated by combining plant and pollinator relative abundances under an independence assumption, whereby the expected probability of interaction between a plant and a pollinator species was proportional to the product of their relative abundances.

Phenology probability matrices were calculated following a similar approach, but abundances were standardized within each species by dividing weekly observations by the total number of observations for that species. This procedure generates a temporal probability distribution describing the likelihood that a species is phenologically active in a given week. Weekly plant and pollinator probabilities were then combined under an independence assumption to estimate temporal interaction probabilities for each plant-pollinator pair.

Trait-based probability matrices were constructed using a size-matching approach between plant and pollinator traits. Pollinator intertegular distance and plant flower width were averaged at the species level to obtain a single representative trait value per species. Trait compatibility for each plant-pollinator pair was quantified using a Gaussian matching function, in which similarity declines smoothly with increasing trait mismatch and is controlled by a tolerance parameter. Pairwise matching scores were assembled into a global plant $\times$ pollinator matrix and subsequently subset for each botanical garden by aligning rows and columns

to the species composition and ordering of each garden’s interaction network and temporal scale.

First, matrix analyses were used to evaluate how similar interaction frequency matrix configurations were across temporal scales for each driver. This was assessed using Procrustes analysis implemented through the *protest* function, and the strength of configuration similarity was quantified using the Procrustes correlation coefficient (r). Analogous analyses were conducted using the Mantel test using the *mantel* function, and the strength of the association was quantified using the absolute values of the Mantel correlation coefficient. Because probability matrices for all drivers were scaled between 0 and 1, Euclidean distance was used as the distance metric in both analyses, as it preserves the geometric relationships among observations compared to alternative distance metrics.

Generalized additive model

To investigate the ecological drivers of plant-pollinator interactions, we modelled the number of interactions recorded for each plant-pollinator pair during each sampling period using generalized additive mixed models (GAMMs). Models were fitted using a negative binomial error distribution with a log link to account for overdispersion in interaction counts. Predictor variables included plant resource availability, pollinator abundance, floral morphology, trait matching between plants and pollinators, and phenological overlap between plant flowering and pollinator activity periods. Trait matching was quantified using a Gaussian function describing morphological compatibility between floral and pollinator traits. We also included interactions between plant resource availability and pollinator abundance, and between floral morphology and pollinator abundance, to test whether the effects of floral resources and morphology depend on pollinator availability. All continuous predictors were standardized prior to analysis to facilitate comparison of effect sizes.

Seasonal variation in interaction dynamics was modelled using garden-specific smooth functions of sampling week, allowing temporal trajectories of visitation to vary among study sites. Repeated observations of the same plant-pollinator pair were accounted for using a random-effect smooth term. Differences in sampling effort were controlled by including an offset equal to the logarithm of the total observation time per plant and sampling week, thereby modelling interaction rates rather than raw interaction counts.

The relative contribution of predictors was evaluated using a drop-one model comparison. Each predictor was sequentially removed from the full model and the resulting change in Akaike’s Information Criterion (ΔAIC) was calculated relative to the full model. Larger increases in AIC indicate a stronger contribution of that predictor to explaining variation in plant-pollinator interaction frequency.

Pollinator preferences

To identify interaction patterns that cannot be explained by plant and pollinator abundances and sampling structure alone, we compared observed interaction frequencies with expectations derived from null models. Interaction data were aggregated at the level of plant and pollinator

families and interaction matrices were constructed separately for each botanical garden and sampling week to preserve the spatiotemporal structure of the data. Within each garden-week matrix, interactions were randomized using a null model that maintains row and column totals, thereby preserving the total number of interactions per plant and pollinator family while randomizing partner identities. For each matrix, 1000 randomized networks were generated. Randomized matrices were then summed across garden-week units to obtain expected interaction frequencies for each plant-pollinator family pair.

Deviations from the null expectation were quantified using standardized effect sizes (SES), calculated as the difference between observed and expected interaction frequencies divided by the standard deviation of the null distribution. Positive SES values indicate associations (more interactions than expected), whereas negative values indicate avoidance. Statistical significance was assessed using two-tailed p-values based on the simulated distributions. Plant-pollinator family pairs that never co-occurred within the same garden and sampling week were considered structurally impossible given the spatiotemporal sampling design and were excluded from the null comparison.

Random forest

Results

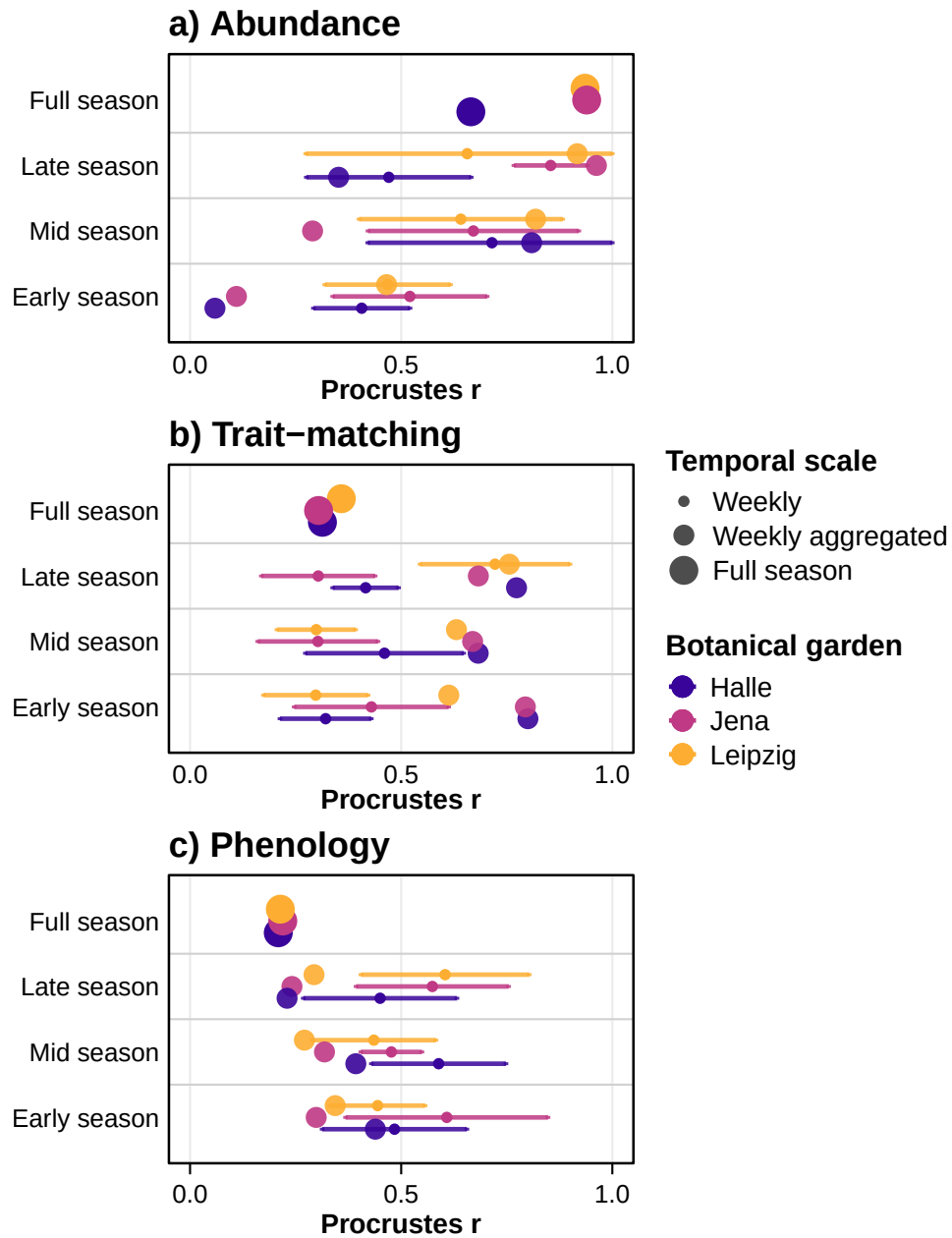


Figure 2. Procrustes association between visitation rate and abundance (a), phenology (b), and traits (size matching; c) across botanical gardens and three levels of temporal complexity (weekly, weekly aggregated and full season). Weekly aggregated values correspond to three seasonal categories (Early, Mid, and Late), obtained by pooling weekly networks into three equal temporal partitions of the full sampling period. Weekly values represent individual weekly networks and are displayed within the corresponding seasonal category. Weekly values are summarized by their mean and 95% confidence intervals for each garden and seasonal category. Full-season values correspond to networks aggregated across the entire sampling period for each botanical garden. Temporal complexity is represented by point size, and botanical garden by colour.

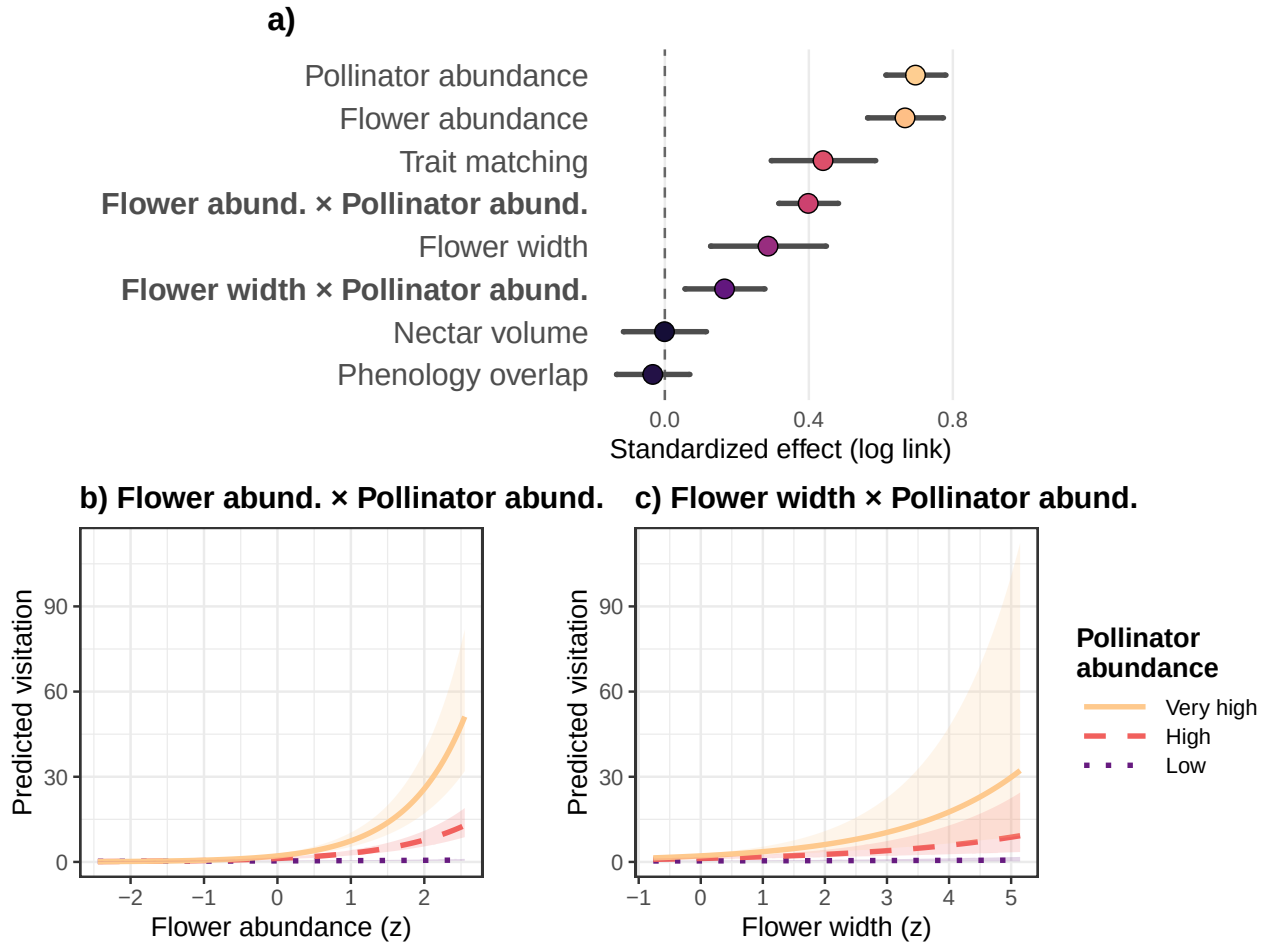


Figure 3. Effects of ecological drivers (i.e., abundance, traits, and phenology) on the number of plant-pollinator interactions. (a) Standardized model coefficients (log link scale) with 95% confidence intervals, illustrating the direction and relative strength of the ecological drivers. (b) Relative importance of ecological drivers expressed as increment of AIC, indicating the change in model fit associated with each predictor. (c-d) Predicted number of plant-pollinator interactions across gradients of flower abundance (c) and flower width. Lines show predicted values from the fitted model and shaded areas represent 95% confidence intervals. Different line types and colour gradients indicate pollinator abundance levels corresponding to the 10th (low), 75th (high), and 90th (very high) percentiles of standardized log pollinator abundance.

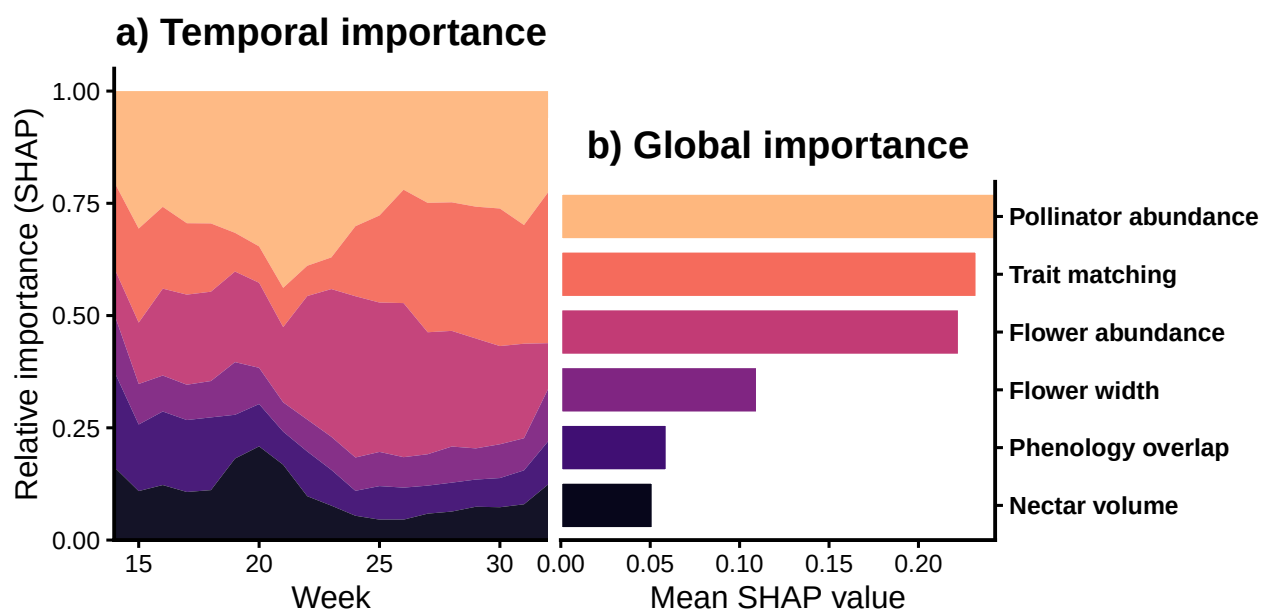


Figure 4.

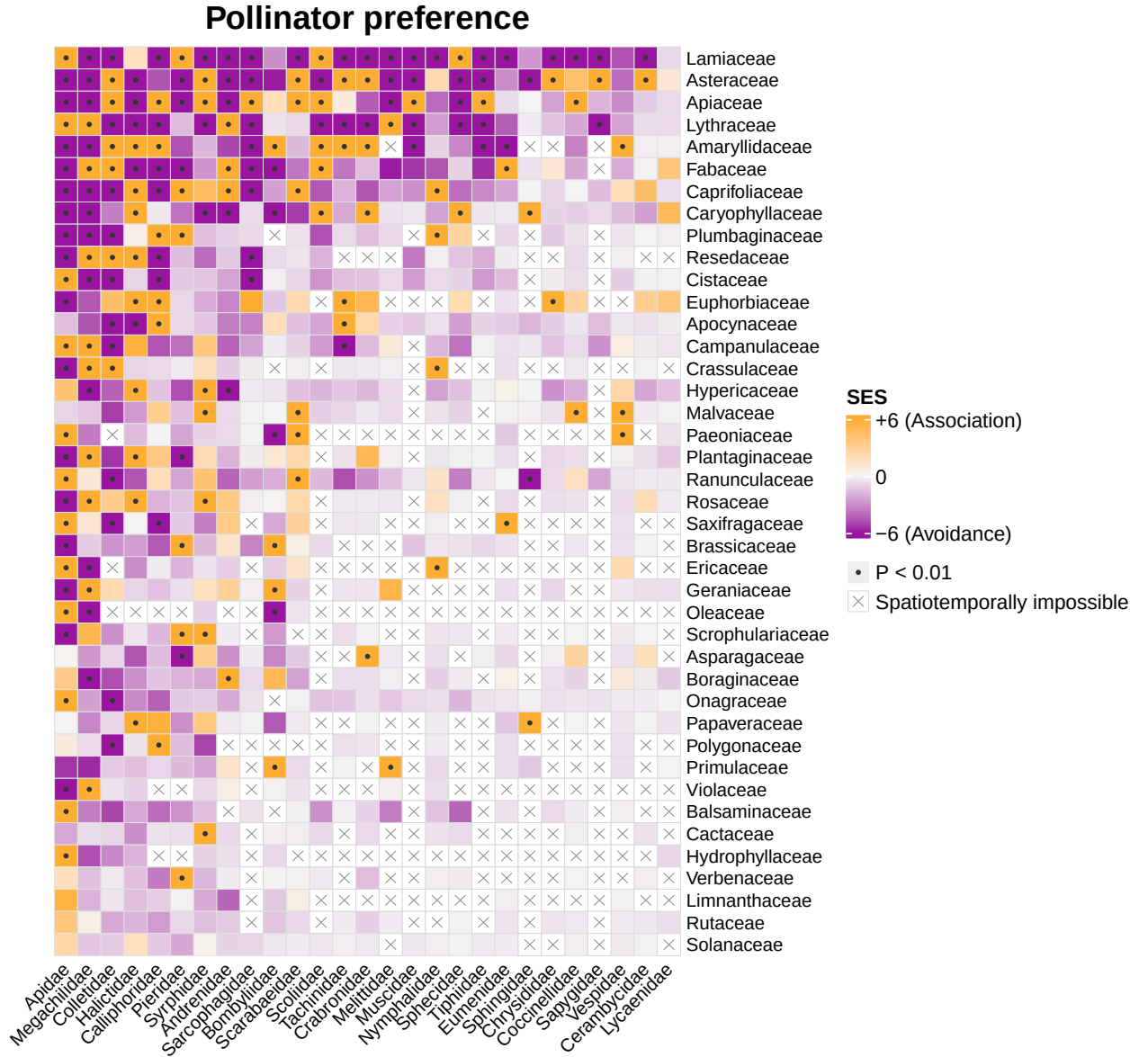


Figure 5. Heatmap of standardized effect sizes (SES) comparing observed plant-pollinator interactions at the family level with expectations from a spatiotemporally constrained null model. Purple indicates avoidance and orange indicates association relative to the null expectation. Black dots denote significant deviations from the null model ($p < 0.01$). Crossed white cells indicate plant-pollinator family pairs that never co-occurred within the same garden and sampling week.

Discussion

Species interactions are inherently dynamic across temporal scales (CaraDonna et al., 2017; Schwarz et al., 2020; Trøjelsgaard & Olesen, 2016), reflecting strong seasonal turnover and within-season shifts in community composition and activity, with corresponding variability in the processes structuring ecological networks (Albor et al., 2020; Delmas et al., 2019; Domínguez-García et al., 2026). By integrating high-resolution data on phenology, traits, and independently measured abundances, our study evaluates how ecological drivers shape species interactions through time. Our results indicate that abundance-based processes dominate interaction patterns, particularly under conditions of high activity (e.g. near phenological peaks), while trait matching refines interaction outcomes, primarily influencing visitation rates rather than overall network structure and becoming more evident under more flexible analytical approaches. In addition, persistent patterns of preference and avoidance across taxonomic levels show that interactions are not solely determined by abundance and trait constraints, but are also shaped by non-random processes such as behavioural preferences or other functional traits. The relative contribution of these drivers shifts through time, yet their effects can be disentangled, allowing the identification of the dominant processes structuring interactions at any given point. Within this framework, our results show that interaction structure is primarily driven by abundance-based processes, consistent with previous evidence identifying species abundances as key determinants of species interaction patterns (Dormann et al., 2017; Fort et al., 2016; He et al., 2025; Simmons et al., 2019). However, this effect is strongly context dependent, as floral resources translate into higher visitation only when pollinators are abundant, with pollinator generalisation increasing during periods of peak phenological overlap. When floral resource availability is explicitly accounted for, apparent specialisation is reduced, indicating that much of the observed generalisation reflects variation in resource availability rather than shifts in interaction preferences. This highlights the need for caution when inferring interaction patterns in the absence of independent abundance data, a limitation that remains common in ecological network studies (Blüthgen & Staab, 2024; Olito & Fox, 2015). Nevertheless, consistent patterns of preference and avoidance indicate that non-random processes, although comparatively weaker, still contribute to structuring species interactions, as also indicated in other studies (?; ?; ?). In contrast, trait matching exerted a consistent, intermediate influence on interactions across temporal scales, suggesting a temporally stable constraint on interactions. While abundance shapes overall interaction patterns, trait compatibility imposes persistent constraints on which interactions can occur (Bartomeus et al., 2016). This is consistent with evidence across systems, including the correspondence between pollinator morphology and floral traits, such as the matching between proboscis length and floral tube depth in Hymenoptera (Klumpers et al., 2019; Zhao et al., 2022), as well as broader macroecological patterns linking hummingbird bills and floral morphology across biomes (?). In our study, size matching (i.e. body size-flower size) was used as a proxy for trait compatibility, as key traits such as tongue length were unavailable for many taxa, often required extrapolation, and showed lower predictive performance than size-based metrics. This aligns with previous work highlighting body size as a strong predictor of interactions, while also suggesting that predictive accuracy improves when multiple traits are considered jointly (Eklöf et al., 2013).

The comparatively weaker role of floral resources is consistent with a hierarchical filtering process, in which trait compatibility constrains feasible interactions, and resource-related factors influence interaction strength only once this constraint is met (?). Phenology, independently derived from plant monitoring and pollinator occurrence records, captures species' activity beyond the limits of the interaction network. While it can constrain feasible interactions by limiting temporal co-occurrence, phenology played a limited role in structuring interactions in this system and weakened with temporal aggregation. This pattern may reflect both the loss of fine-scale temporal structure through aggregation (CaraDonna et al., 2017; Schwarz et al., 2020), and the high floral resource availability and broad temporal overlap observed in this system, which can relax phenological constraints and increase the range of feasible interactions (Benadi et al., 2014; Manincor et al., 2020). Such conditions may be particularly pronounced in urban or human-modified systems, where extended flowering periods can buffer seasonal declines and maintain relatively stable pollinator richness across the season (Leong et al., 2016). By contrast, in more resource-limited or phenologically constrained systems, phenology plays a stronger role in structuring interactions and fitness outcomes (Encinas-Viso et al., 2012; Vázquez et al., 2023), and may provide stronger predictive power across seasons than within-season interaction dynamics (Bramon Mora et al., 2020; Peralta et al., 2020; ?). Importantly, the influence of these drivers is not constant through time. Instead, their relative importance shifts across the season, with encounter-driven processes dominating during periods of high activity (e.g. near phenological peaks), and trait-based or resource-related constraints becoming more influential as encounter rates decline. This highlights that interaction structure emerges from temporally dynamic processes rather than fixed mechanisms (?; ?), with different drivers prevailing under different ecological conditions. Despite this variability, a relatively small set of ecological drivers captures much of the variation in visitation patterns, as reflected by the moderate to high predictive performance of the models, supporting the view that complex ecological dynamics can be predicted from a limited set of key drivers (Peralta et al., 2024; Pichler et al., 2020). This suggests that interaction dynamics can be understood through a combination of dominant processes and context-dependent constraints. Together, these patterns support a general framework in which abundance governs interaction frequencies, while traits and phenology constrain which interactions are feasible, with their relative influence varying across temporal contexts. Although intraspecific variability can influence plant-pollinator interactions (Arroyo-Correa et al., 2021; Kuppler et al., 2016), the low genetic diversity and frequent clonality of plants in botanical gardens likely reduce the role of individual variation in this system. The comparatively lower importance of trait matching inferred by GAMs relative to random forests further suggests that conventional modelling approaches may underestimate its contribution, particularly when interactions are structured by complex or multiple trait combinations (Pichler et al., 2020). Beyond within-season dynamics, interactions can also rewire across years (Domínguez-García et al., 2026; Resasco et al., 2021), highlighting the importance of incorporating interannual variability and assessing how climatic fluctuations influence the relative importance of interaction drivers. More broadly, the processes structuring species interactions are likely to be context-dependent. For example, alpine communities may be more strongly constrained by phenological dynamics (Kudo & Cooper, 2019), whereas tropical systems may be less limited by abiotic constraints and

therefore more shaped by niche-based processes (Schemske et al., 2009). Similarly, the relative importance of mechanisms may vary across taxonomic groups, with bee-pollinated systems often showing a stronger influence of abundance-based processes, whereas hawkmoth-plant (Sazatornil et al., 2016) and hummingbird-plant networks (?) tend to be more structured by trait matching. Together, these patterns highlight that the drivers of ecological interactions are inherently context-dependent, requiring integrative and temporally explicit approaches to fully understand their dynamics. Overall, our results show that plant-pollinator interactions emerge from a limited set of ecological drivers whose relative importance shifts through time. While abundance governs encounter rates, traits and phenology constrain which interactions are feasible, generating dynamic but structured interaction patterns. By integrating high-resolution temporal data with independent estimates of abundance and complementary analytical approaches, our study demonstrates that these processes can be disentangled and quantified, even in highly dynamic systems. This provides a general framework for understanding species interactions, and highlights the need to incorporate temporal dynamics and independent ecological information to better assess how ecological networks may respond to environmental change.

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Supplementary information

Table S1 Summary of plant and pollinator traits

Traits	Mean	SD	Min	Max
a) Plant traits				
Autonomous selfing	6.45	21.67	0.00	88.00
Corolla diameter	26.22	22.09	4.50	115.00
Flower lifespan	8.08	8.38	1.00	48.00
Flowering length	50.48	26.62	10.00	129.00
Flowers per plant	550.97	1406.11	1.00	8000.00
Nectar volume	1.33	3.80	0.00	23.25
Ovule number	66.68	92.00	1.00	390.00
Plant height	0.53	0.33	0.04	1.49
Pollen per flower	381413.84	1461347.93	73.50	11994432.00
Pollen size	42.10	28.81	17.50	240.00
Seed mass	0.01	0.04	0.00	0.35
Style length	9.48	13.25	0.00	84.85
b) Pollinator traits				
Body length	9.04	3.56	2.07	23.98
IT	3.05	1.37	1.18	9.13
Proboscis length	3.92	3.13	0.40	17.91

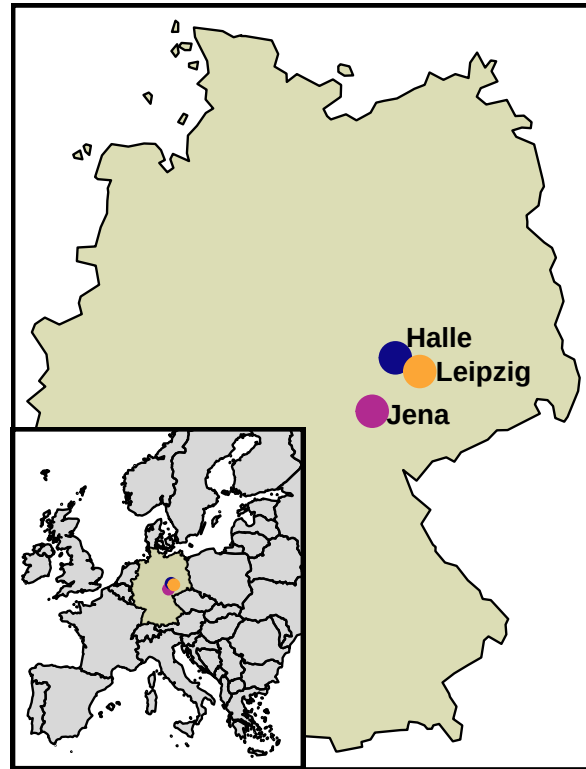


Figure S1.Map showing the locations of the three botanical gardens included in the study.

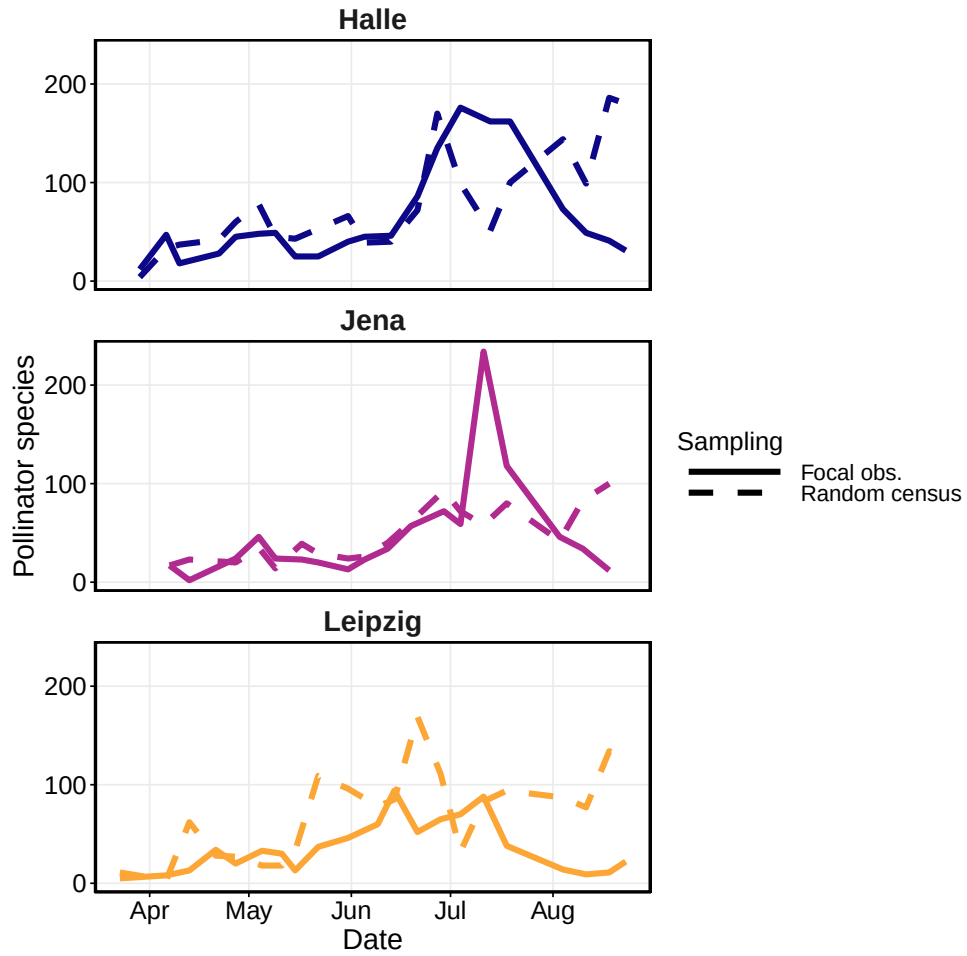


Figure S2. Temporal dynamics of plant-pollinator interactions across three botanical gardens. Solid lines represent focal observations on PhenObs plants, whereas dashed lines indicate random census observations across the gardens. Each panel corresponds to a different botanical garden: (a) Halle, (b) Leipzig, and (c) Jena.

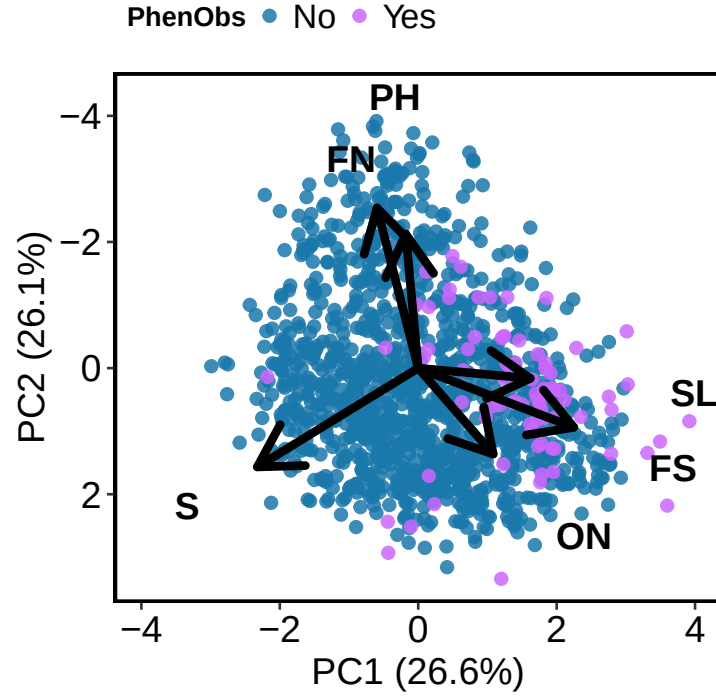


Figure S3. Position of focal plant species within the global angiosperm reproductive trait spectrum. Blue points represent species from the global dataset of floral reproductive traits described by Lanuza et al 2023. Purple points correspond to the PhenObs species monitored in this study. Arrows indicate the loadings of the main reproductive traits on the first two principal components. The focal species cluster within the region of the spectrum associated with higher investment in floral traits linked to pollinator attraction, consistent with the focus on pollinator-dependent taxa in this study. This figure illustrates the position of the sampled taxa within the broader reproductive trait space rather than representing a formal analysis.

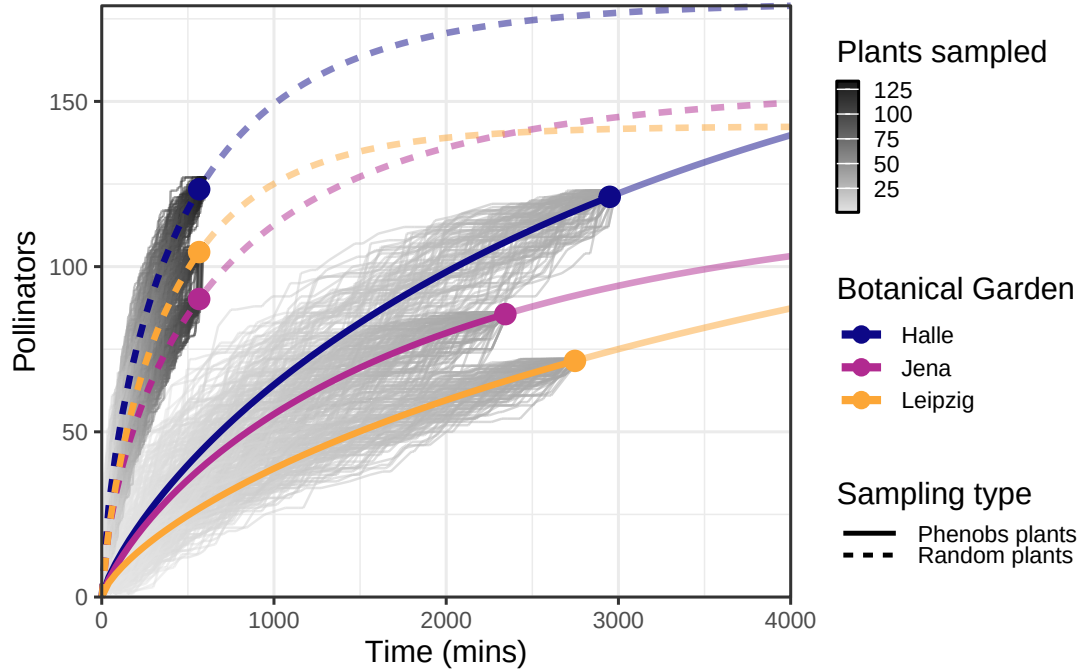


Figure S4. Accumulation curves of pollinator species across botanical gardens and sampling methods (dashed and solid lines). Each grey line represents one iteration of the accumulation curve, with a total of 100 iterations per garden and sampling method. The coloured lines represent the rarefied mean accumulation curves, with the dot indicating the total number of observed species. The segments of the lines extending beyond the dot represent the extrapolated number of expected species.

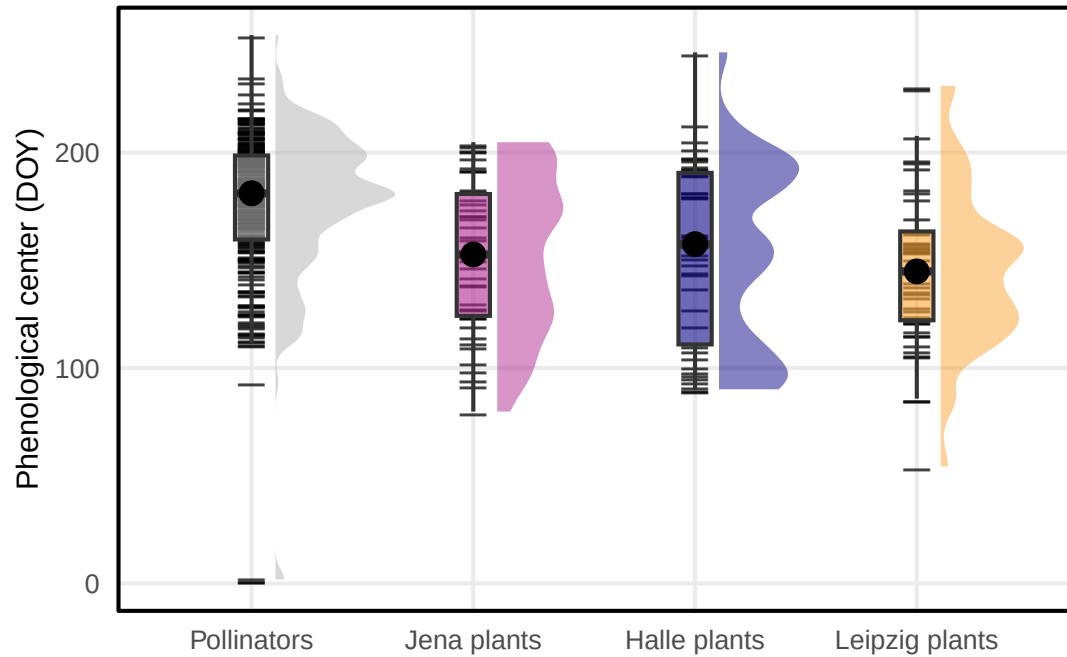


Figure S5. Phenological centres of flowering plants and pollinators across botanical gardens. Distributions of phenological centres (day of year, DOY) are shown for pollinators and flowering plants in Jena, Halle and Leipzig. Pollinator phenology was estimated at the regional level from occurrence records and is therefore represented as a single distribution. Half-eye densities and boxplots show the distribution and spread of values, with black points indicating medians and horizontal line segments representing individual observations. Pollinators peaked later in the season than plants.

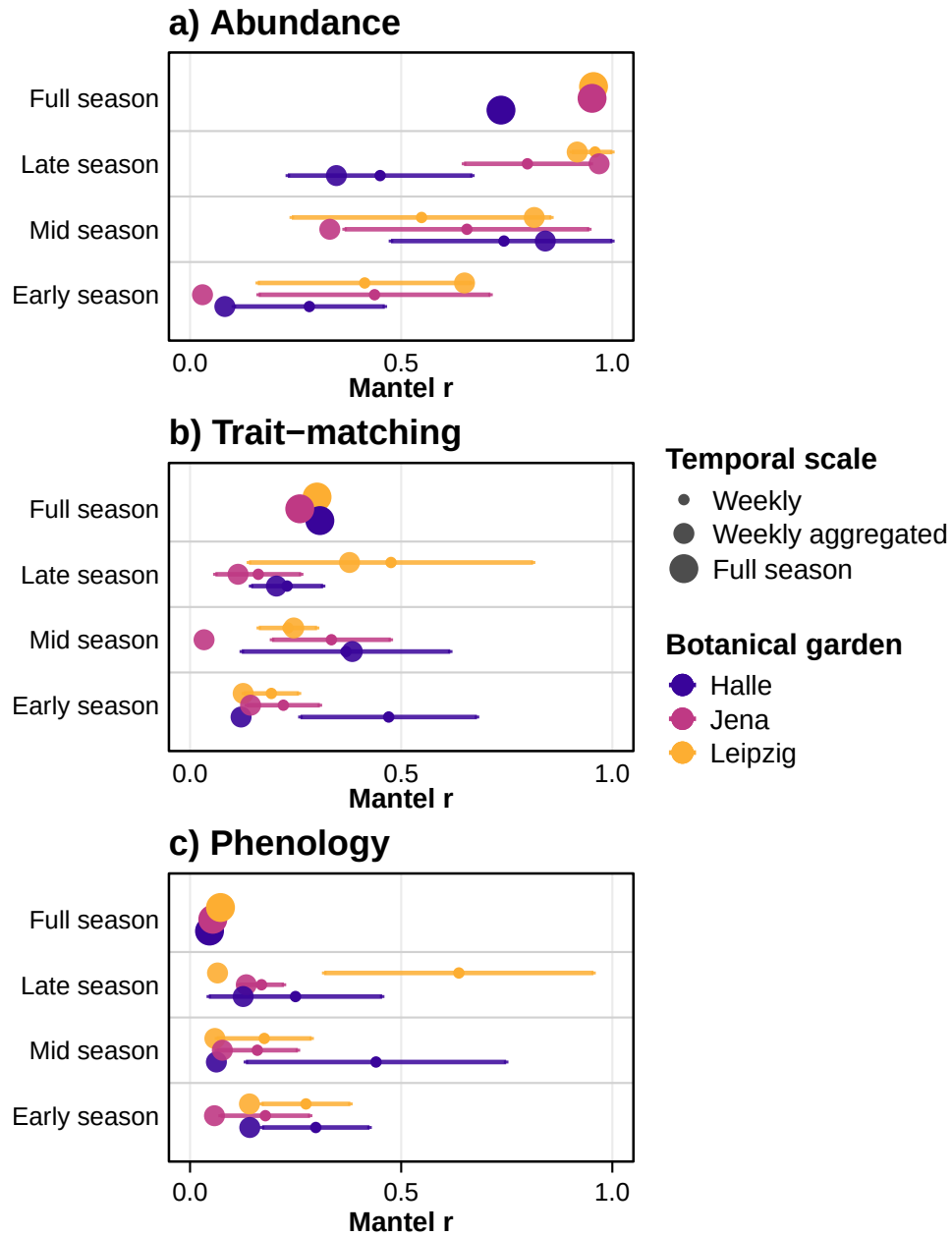


Figure S6. Mantel correlation between visitation rate and abundance (a), phenology (b), and traits (size matching; c) across botanical gardens and three levels of temporal complexity (weekly, weekly aggregated and full season). Weekly aggregated values correspond to three seasonal categories (Early, Mid, and Late), obtained by pooling weekly networks into three equal temporal partitions of the full sampling period. Weekly values represent individual weekly networks and are displayed within the corresponding seasonal category. Because multiple weekly values are available per garden and seasonal category, weekly data are shown as individual points together with their mean and 95% confidence intervals (± 1.96 SE). Full-season values correspond to networks aggregated across the entire sampling period for each botanical garden. Temporal complexity is represented by point size, and botanical garden by colour.

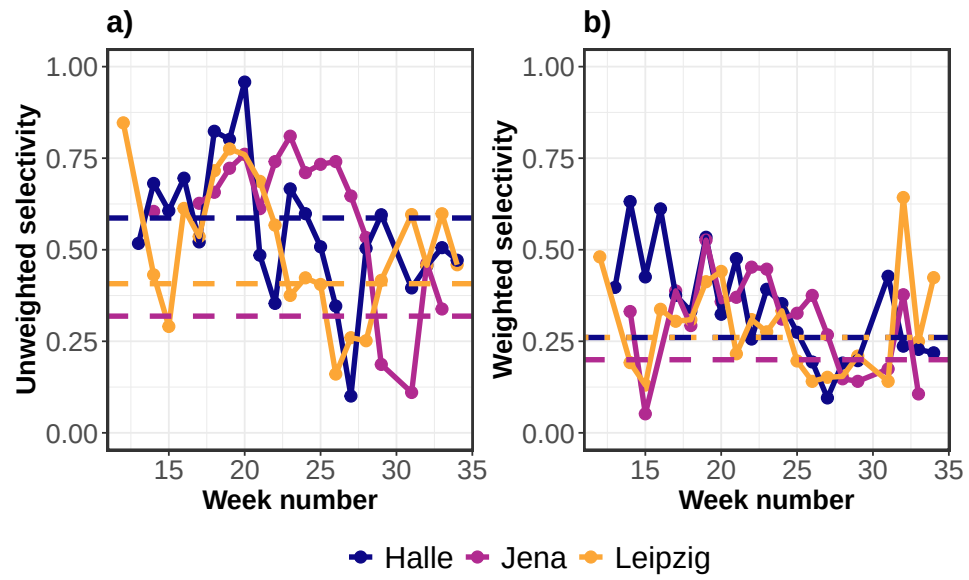


Figure S7. Temporal variation in mean pollinator selectivity across botanical gardens. Weekly mean selectivity (d') was calculated from interaction frequency matrices (a) and after accounting for floral resource availability (b).

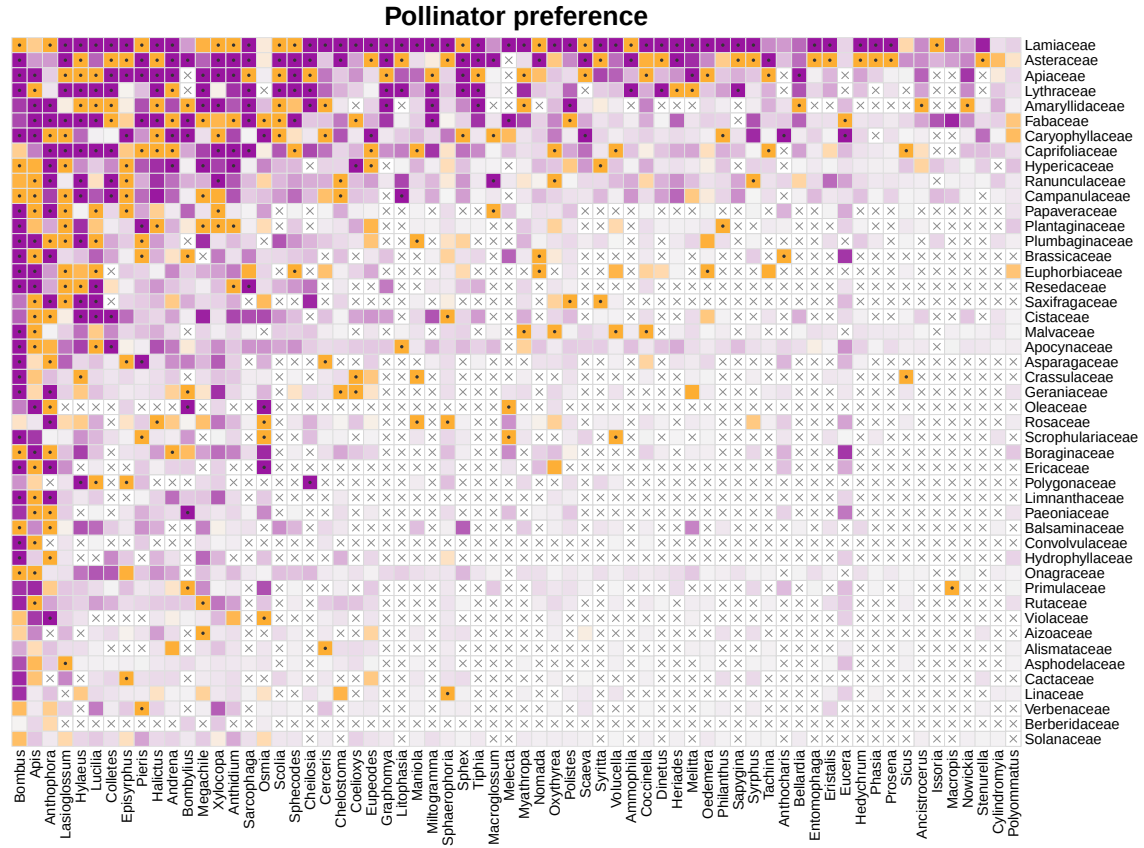


Figure S8. Heatmap of standardized effect sizes (SES) comparing observed plant-pollinator interactions at the family level with expectations from a spatiotemporally constrained null model. Purple indicates avoidance and orange indicates association relative to the null expectation. Black dots denote significant deviations from the null model ($p < 0.01$). Crossed white cells indicate plant-pollinator family pairs that never co-occurred within the same garden and sampling week.